

CSE 578 Final Project Personal Reflection

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1 INTRODUCTION

In my data visualizations course CSE 578, my final project was a group project asking students to pick a controversial topic and create two scrollytelling web pages showcasing opposite perspectives on the topic. The scrollytelling stories had to use D3, a Javascript library used to create data visualizations, HTML, CSS, and Javascript in the creation of twelve visualizations, six for each side, which supported the respective arguments being made and were in line with compelling prose which explained each argument. I worked with Jessica Jones, Aditya Kumar, DeJuan Laury, Nicholas Nabours, and Brennan Sammis to complete this project.

2 SOLUTION

When choosing a topic, the group decided early on that we wanted to work on something lighthearted that still had a lot of data to pull from. We settled on the age-old debate of whether cats or dogs were the better pet. Like the project description detailed, we split into two groups of three: Aditya, Nicholas, and Brennan worked on the cat argument and visualizations while Jessica, DeJuan, and I focused on the dog side of the debate. This distinction was mostly in-name only, as the group helped each other to complete both sides of the topic throughout the project. The group worked together to brainstorm the visualizations, which needed to include six unique techniques and two innovative visualizations. The initial plan was for members of the group to complete a particular visualization style that could be reused by both sides in order to streamline the workload. However, all group members ended up making two visualizations due to the needs of the datasets each member was visualizing. Many datasets the team used were sourced through Kaggle, an online platform where people share datasets and machine learning models [1]. Others were found through groups such as the American Veterinary Medical Association (AVMA), which publishes many “clinical and research studies in veterinary medicine”.

3 RESULTS

3.1 My Results

3.1.1 Normalized Stacked Bar Chart. The first visualization I created was a normalized stacked bar chart. This technique is a variant of the regular bar chart that normalizes data of a particular category before separating each bar totaling 100% into its quantitative parts within the category. This makes the technique one with high information density, as more pixels within the

visualization are displaying information to the viewer than the average chart. This visualization featured data from a paper published by the AVMA exploring emergency department (ED) visits for people who suffered from dog and cat bites in California from 2005 - 2019 [2]. The data I took was from a table distinguishing emergency department visits for dog bites and cat bites and binning, or separating, this data into age groups. Young ages were binned in five year increments, while visits including people twenty and over were binned in ten year increments. After loading the dataset into javascript, I normalized each age group by totaling the bites in each category and dividing each animal's bite statistic by this total, thereby making the columns percentages of each row's total. When plotting the data, it was evident that dog bites resulted in more emergency department visits than cat bites, so it made much more sense to implement this visualization on the cat's side.

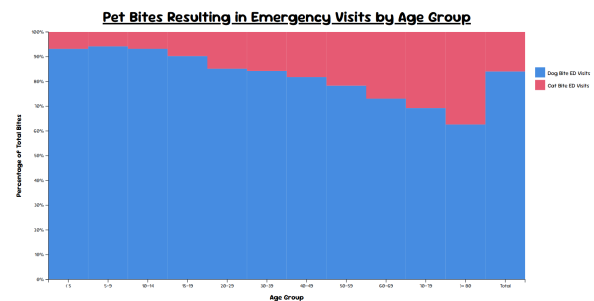


Figure 1: The normalized stacked bar chart created using D3 displaying the disparity in ED visits from animal bites. Blue represents dogs while pink represents cats.

After creating a static version of the visualization, I added transitions using joins and an enter, update, exit structure. This structure streamlines the implementation of transitions integral to the scrollytelling medium. When the visualization is scrolled on top of, all parts of the visualization fade into view and the bars grow downward from the top of the visualization. When scrolling away, the bars grow back up and all parts of the chart fade out. This visualization was not custom, but it displayed its message clearly, supporting the cat argument well.

3.1.2 Combinatorial Choropleth Map. The second visualization I developed was a choropleth map of the United States (US) overlaid with radar charts on top of each state. Choropleth maps are a common visualization technique in which the data is usually presented by coloring geographic areas on a map based on their data values. One of the biggest issues with choropleth maps is the “don’t forget about Rhode Island” problem, in which

small geographic areas are unseeable or lost due to them being small in relation to the total area shown on the map. This problem was especially important to consider when overlaying the states with radar charts. Radar charts are another form of bar chart in which a polygon represents values on radially positioned axes. Each axis around the circle corresponds to a categorical attribute, and the polygon maps quantitative values to the axes, creating an interesting polygon. The choropleth was made using AVMA data from 2016 looking at pet dog and cat ownership rates [3]. The dataset showed the percentage of households which owned a particular pet in each state, and I used these percentages to derive an attribute of the difference in dog and cat ownership in each state. This attribute's data was plotted using a diverging color scale on each of the forty-eight continental states since the dataset did not include Alaska or Hawaii. The map of the US itself was built using TopoJSON, an extension of GeoJSON that "eliminates redundancy [by] allowing related geometries to be stored efficiently in the same file," which was appealing for its use in a big project and a visualization that needed to be loaded quickly and possibly multiple times [4]. Next, I created a dataset from personality data collected and visualized by NERIS Analytics Limited, a research group focused on the Myers-Briggs personality types [5]. I noted the reported percentage difference for each personality trait in each state, then charted these percentages against their trait counterpart on the radar chart. A radar chart was constructed for each of the forty-eight continental United States. To respect the "don't forget about Rhode Island" problem, the some radar charts in the northeast were offset from their state slightly to ensure their choropleth encoding was still visible alongside the radar chart. The datasets together show correlations between personality traits and the pets people choose to own. Alongside the map, legends detailing the diverging color scale and the radar chart axes were created and placed appropriately within the visualization.

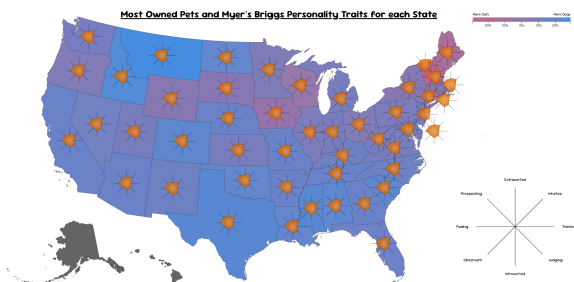


Figure 2: *The combinatorial choropleth map with radar charts overlaid created using D3 displaying the differences in pet ownership by state in 2016 and correlating this information with personality trait data*

For the choropleth map, I created staggered transitions to make the charts build and dissipate in stages rather than all at once. When the user scrolls to the visualization, the map fades in, then the area encodings. Afterwards, the radar charts drop onto each state on the map, and finally, the legends then the title and a descriptive paragraph fade in. The elements fade out in the opposite order when the user scrolls away. Each element used the same enter, update, exit methodology. This visualization would be considered innovative. In visualization terms, a chart can be described as innovative if it is modded, combinatorial, or novel. Modded charts mainly focus on adding interaction or graphical elements to a well-known chart. Combinatorial charts integrate multiple visualization techniques. Novel charts are usually created in relation to a single use case and display data in a unique way, whether for better or for worse [6]. This visualization clearly falls into the combinatorial category as I overlaid radar charts over states on a choropleth map, combining two distinct chart types.

3.2 Integration

As the team completed their visualizations, they were integrated into one set of files. Nicholas developed the structure for the main web page using Scrollama, a library that helped streamline the scrollytelling transitions necessary for the project [7]. Some visualizations took longer to create than others, and some were added with bugs their creator could not fix. As I finished my static charts relatively early, I had a lot of time to help find bugs in other people's charts and add the transitions to my own. An interesting bug I helped fix was with Aditya's Sun Chart. When the chart opened as the user scrolled to it, if the user moused over the polygons while they were settling, their shape would set as an interactive tooltip was displayed. This caused their shape to be warped permanently until the chart was remade by scrolling away and then back to the chart. I fixed this bug by disabling events triggered by mousing over the polygons while they were gaining their shape and settling in place. Once the polygons were set, the events were re-enabled. Another visual issue I resolved regarding the polygon's interaction was how the selected polygon was made visually distinct. When interacted with, polygons would attempt to make their border black, but this was variably applied around the polygon, causing the retention of white edges. I replaced this with a brightening effect on the polygon and a slight drop shadow, making it visually distinct without editing the polygon's border. I helped with other bug fixes on charts throughout the integration process. The other main aspect of integration that I was a part of was standardization. This involved making all the chart's fonts and colors the same as one another to keep a consistent look throughout each side's story. This

required parsing other team member's code to find where colors or fonts needed to be set. This effort also included centering and resizing charts as well as making simple transitions in relation to scrolling for charts that had none created. The standardization felt worthwhile in the end as the project looked more appealing as a whole.

3.3 Story

The story was the last element of the project to be finalized. It took several attempts to get the visualizations in the order the group felt was natural and then write a story tying the charts together. While my main focus on the team was the development of the web pages, I did understand, advise on, and contribute to the story.

3.3.1 Cat Argument. For the cat's side, the argument had three main points. The first point argued that cats were better due to their stronger online presence. Data shown through a box-and-whisker plot and custom pyramid visualization showed that online communities centered around cats are bigger than communities focused on dogs. The second point was about health and safety. My normalized stacked bar chart showed that people were more likely to visit emergency departments for dog bites than for cat bites, and a line chart emphasized that dog bites cause more cases of rabies each year than cat bites. Finally, cost was used as an argument for why cats are better than dogs. In initial one-time expenses, first year costs, and annual expenses, cats are much cheaper to own than dogs. The cost data was displayed in a diverging bar chart that transitioned as scrolling occurred. These main arguments were also supplemented by a counterargument regarding cat diversity. A sun chart was used to show the diversity of cat breeds and how many there are to choose from, a fact that I hadn't thought about before this project.

3.3.2 Dog Argument. The dog's argument was less cohesive than the cat's argument, but still strong. First, a line chart and my choropleth were used to show the overwhelming difference in the percentage of dog ownership and cat ownership in favor of dogs. The choropleth map tied in the fact that consistent personality traits across states with high dog ownership showed a correlation with common personality traits, concluding that most people are dog people. This shifted to survey data our group collected from our class which showed the class believed that dogs contribute more to humanity. This fact from the survey was expounded upon with governmental and medical data showing the types of jobs dogs do to help humanity. Stories of dogs helping humans were also told to bolster this point. The point that the variety in dog breeds being greater than cat breeds with another sun chart was made before rounding out the dog argument with the fact that twice the number of people are allergic to cats as to dogs.

4 DISCUSSION

This project gave me a new perspective on how visualizations are created and concrete experience developing interesting and informative charts. Previous projects I have done with respect to data visualization only focused on displaying the numbers within charts; by tying the visualizations directly to a controversial topic, they held a greater purpose in driving a narrative and argument. The project's requirements reinforced the message that data is personal, a theme that, looking back, runs through all of CSE 578. By the end of the project, I felt confident in my ability to develop charts using D3. The creation of the choropleth map improved my understanding of the development patterns of the library greatly as I needed to overlay the radar charts. Once this chart was integrated into the greater project, I had the opportunity to develop and tune transitions for when users scrolled on to and away from the map. This project was also the most intensive web development project I have been a part of, and I became more comfortable using each language and interfacing with the architecture of web pages throughout the project. The most valuable part of this project for me, though, was getting experience parsing other developers' code to patch bugs and standardize formatting. When working anywhere in the tech industry, more often than not I will be continuing development on an already existing system. This reality makes any opportunity to read, use, and debug other people's work invaluable. This project gave me opportunities to refine soft and hard skills that I will utilize throughout my software development career.

5 REFERENCES

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